

Yuan Jack Yao

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EDUCATION

Georgia Institute of Technology

Aug 2025 – Dec 2026

M.Sc. Analytics @ School of Computational Science and Engineering

Atlanta, Georgia

- **Graduate Teaching Assistant @ CS 6604 Conversational AI**
- Courses: Conversational AI, Natural Language, Machine Learning, ML in Computational Biology

National Taiwan University

Sep 2014 – Jun 2018

Bachelor's Degree in Economics

Taipei, Taiwan

WORK EXPERIENCES

Walmart

Jun 2026 – Aug 2026

Data Scientist III Intern @ Sam's Club AI Acceleration Team

Bentonville, Arkansas

- Integrated **LLM & RAG with filtered vector search** into dashboard for answering camera metadata queries at camera/area/site levels, with **cosine similarity & LLMs-as-a-Judge as quality gates against hallucinations**
- Designed an **embedding signature to verify Milvus vector database** ingestion & retrieval sides call the identical embedding model — despite being forced onto different API clients — **eliminating silent retrieval degradation**
- Deployed **monthly Gemini Vision CronJob** across **1,245 in-store cameras** spanning 9 sites to classify snapshot content and map camera to its actual in-store area, with SVG floor layouts as primary reference and CSV as fallback
- Achieved full-fleet **inference of 1,245 cameras in 13 minutes through concurrency**, while **capping glibc malloc arenas and pre-downsampling images to eliminate out-of-memory failures**
- Deployed **hourly CronJob with Docker & Kubernetes**, integrating **asynchronous processing** that reduced **50%** execution time for capturing **1,245 camera snapshots into Azure**, using 7-day TTL to save **75%** storage cost

SUEZ

May 2022 – Sep 2024

Data Scientist @ Smart and Environmental Solutions (SES) China

Shanghai, China

- Streamlined **NLP-based Keras classifier** for **imbalanced SCADA alarms**, reaching **82% production accuracy**
- Designed a **DBSCAN-based anomaly detector** for air compressors **predictive maintenance**, achieving **70% production F1 score** and forecasting failures **two weeks in advance despite having only a single feature**

PERSONAL PROJECTS

LinesShines

- Designed an **interactive system** for exploring NFL players — supports **pinned multi-merging** comparison and embedded teammate-strength summaries, inspired by GitHub's pinned repos & pull request merge mechanism
- Built **draggable rank & percentile panels** that recompute live on **multi team & player union filters**

MonkeyJOB: Jack's Online Blackjack

- Developed **full-stack online Blackjack game** with casino-style UI via FastAPI backend & vanilla JavaScript
- Implemented Macau-style split logic supporting up to 4 concurrent hands with per-hand bet tracking and conditional double-after-split eligibility, alongside no-hole-card insurance and early-pay rules

SkyHorse

- Authored **bilingual tech blog in English and Chinese for 30+ LeetCode hard problems**, delivering detailed tutorials to cover optimized Python & C++ implementations and complexity analysis

OPEN SOURCE

Code Puppy (795 Stars, 267 Forks) — AI Coding Agents

- Audited incorrect automated dead-code report via independent verification of static grep + IDE semantic analysis, removed dead functions and extracted common code, **achieving DRY and preventing confusions**

LeetCode Wiki (36.5k Stars, 9.5k Forks) — DSA Library

- Authored **8 optimal time and space solutions with tutorials** for hard problems
- **No.2454 Next Greater Element IV** optimization: achieved $O(n)$ time complexity by **double stacks**

TECHNICAL SKILLS

Gen AI & ML: AI Infra & Platform, RAG, LLM Fine-tuning, Distillation, PyTorch, Hugging Face, TensorFlow, Keras

CI/CD & Cloud: Linux, GitHub, Bitbucket, Docker, Kubernetes, Azure, AWS, GCP

Programming & Databases: Python, C++, Milvus, Cassandra, MySQL, PostgreSQL, ETL Pipelines

Web & Visualization: FastAPI, Flask, JavaScript, HTML, CSS, Web Scraping